2026 AUTONOMOUS AERIAL VEHICLE CHALLENGE (AAVC2026)

Rules and Regulations

(Version 1.3 – July 2026)

Introduction

The autonomous aerial vehicle challenge (AAVC) is an undergraduate-level event established by the collaborations between aeronautical/aerospace engineering faculties in the Thailand's Consortium of Aerospace Engineering (CASE) since 1997. The event aims for encouraging undergraduate students (focusing primarily, but not exclusively, for aerospace engineering majors), as the teams, to demonstrate their ability to implement both engineering (aeronautical, mechanical, and electrical, and computer, etc.) knowledges and operational expertise to develop, build, and operate their own autonomous aerial vehicles system (known as unmanned aircraft system UAS) to complete the task(s) which simulated based on real-world problem. The participants shall demonstrate their engineering, problem-solving, and critical thinking skills as well as their creativity in developing the system that meets the operational requirements with sufficient safety considerations. The participants will then demonstrate their teamwork, and decision-making in operating their developed system to complete the tasks effectively under specific rules and restrictions with operation safety in mind. The participating teams shall provide a presentation on any aspect of the team's technical achievements during the development of their system. This leads to additional team members being allowed to participate as presenters (see registration details).

In this year The AAVC 2026 will be held from 28-30 August 2026 at the International Academy of Aviation Industry (IAAI), King Mongkut's Institute of Technology Ladkrabang (KMITL), Thailand.

Operating Environment Information: Key Locations, Areas, and Scenario Set-up

The operation challenge scenario will be set over the IAAI's perimeter, with the controlled airspace and search area established as illustrated in Figure 1.

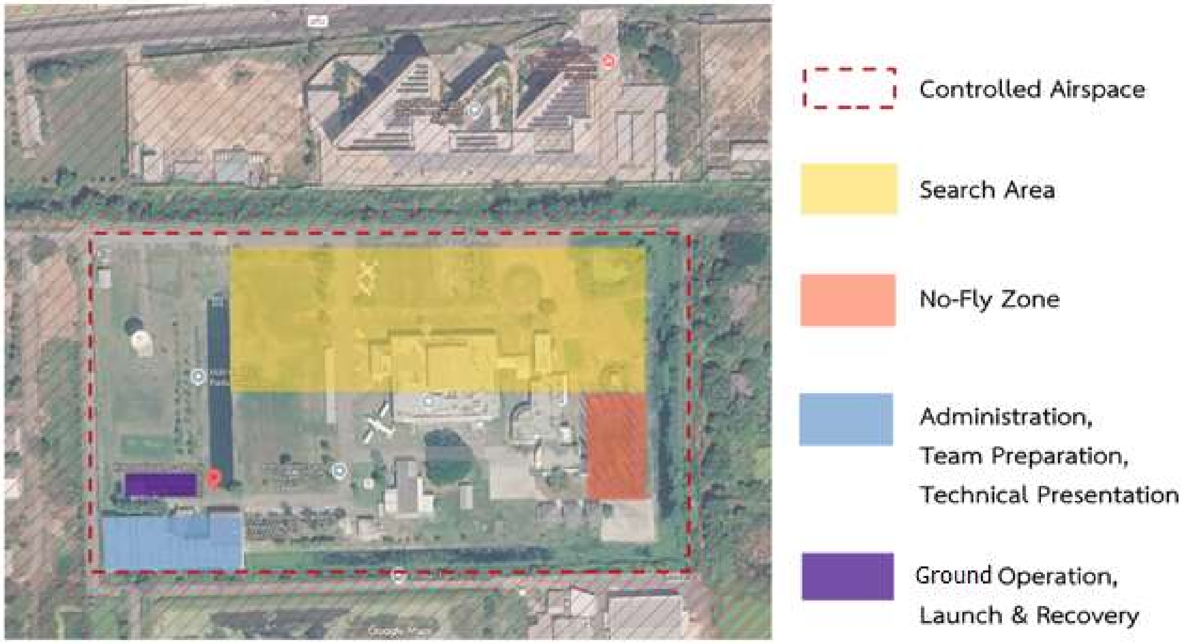

Figure 1. Overlay of the controlled airspace and search area over the IAAI, KMITL

The controlled airspace (acting as a Geo-fencing boundary) is defined using the following coordinates:

13.731312, 100.787175 (13°43'52.72"N,100°47'13.83E")
13.731359, 100.789916 (13°43'52.89"N,100°47'23.70"E)
13.729806, 100.787228 (13°43'47.30"N,100°47'14.02"E)
13.729994,100.789841 (13°43'47.98"N,100°47'23.43"E)

Similarly, the search area is defined by the following coordinates:

13.731239, 100.787824 (13° 43'52.46"N,100°47'16.16"E)
13.731359, 100.789916 (13°43'52.89" N,100°47'23.69"E)
13.730723, 100.787840 (13°43'50.60" N,100°47'16.22"E)
13.730703, 100.789776 (13°43'50.53" N,100°47'23.19"E)

The key ground operation areas within the IAAI's perimeter, including the launch and recovery area, the ground control station operation area, and the stand-by team preparation area, are illustrated in Figure 2.

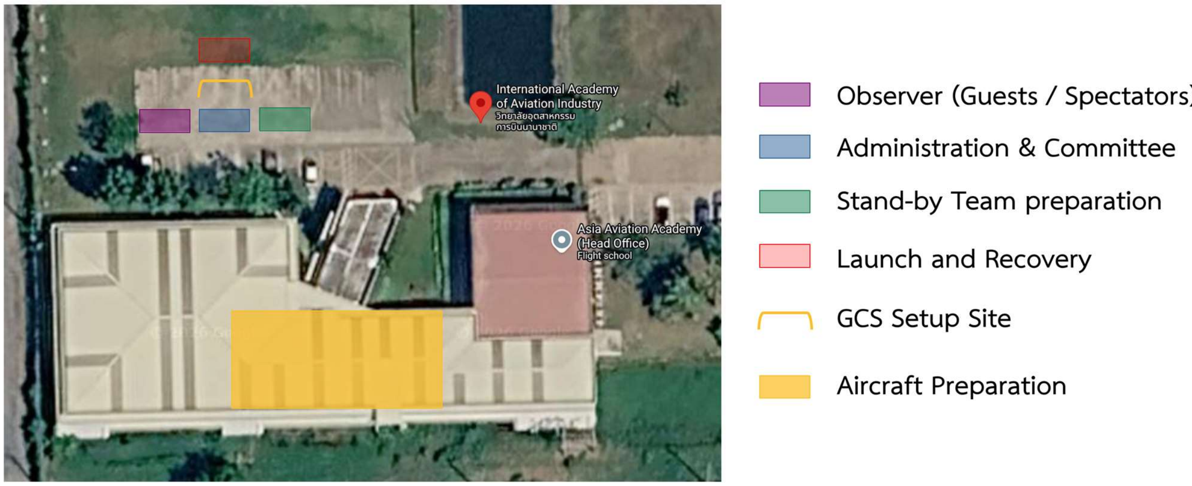

Figure 2. Key ground activity areas over IAAI's perimeter.

The transit flight routes for entering (ingress) and exiting (egress) from the search area are set to ensure flight operations' safety upon entering and exiting the search area within the controlled airspace are illustrated in figure 3. All vehicles must use this transit route to enter the search area after successfully taking off, and to return from the search area before landing. The function of each point in the transit route is described in Table 1.

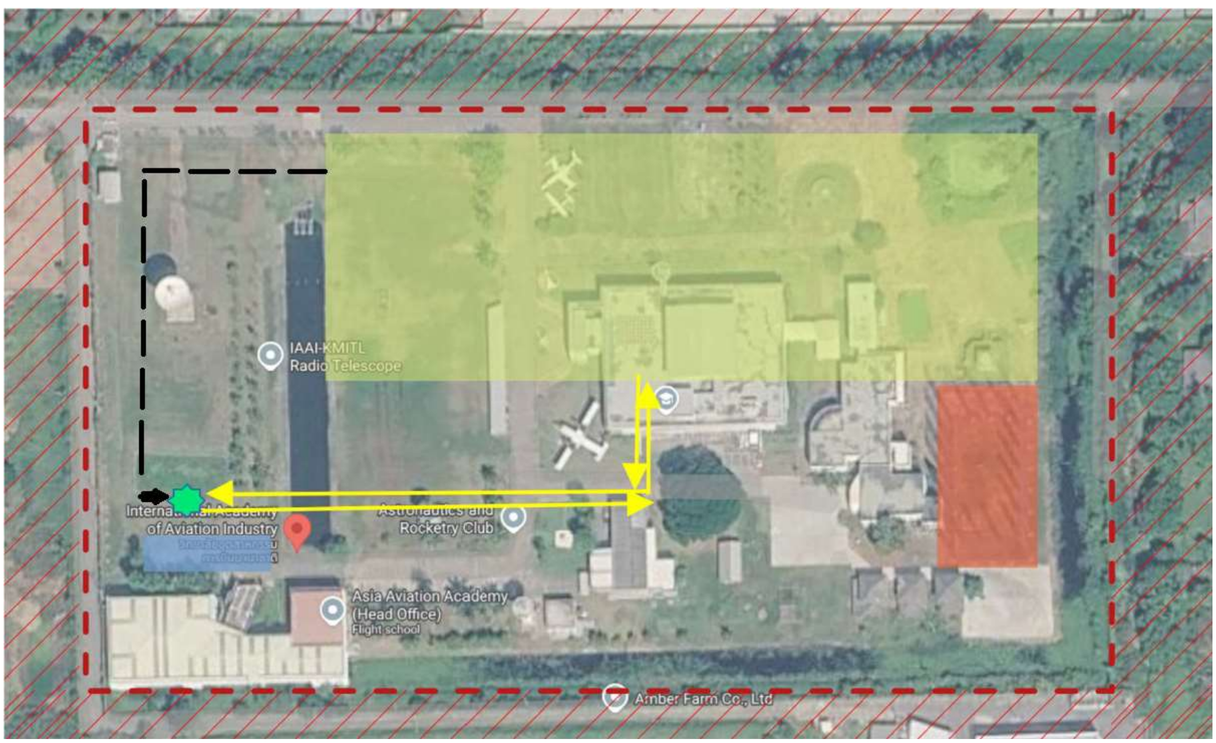

Figure 3. Primary transit route (yellow) and emergency egress route (black).

Table 1. Primary transit route: Waypoints coordinate and functions.

Primary transit route		
Point ID	**Coordinates**	**Function**
1	13.730322, 100.787446 (13°43'49.16"N,100°47'14.80"E)	- Initial transit point after takeoff. - Final transit point before landing. - Return-Home (RTH) / Holding point for system check.
2	13.730397, 100.788694 (13° 43' 49.43"N,100°47'19.30"E)	Intermediate point
3	13.730712, 100.788755 (13°43'50.5632"N,100°47'19.52"E)	Search area ingress/egress point

Recommended Telemetry Frequency Ranges

It is strongly recommended that the participating teams select the telemetry hardware that falls into the available frequency ranges stated below to minimize the complications regarding radio frequency regulation.

Frequency range 1 : 920-925 MHz
Frequency range 2 : 2400-2500 MHz
Frequency range 3 : 5725-5850 MHz

The use of 4G LTE (i.e., a SIM card) is *not allowed* by the Thai authorities due to security reasons. If using the telemetry component with an operating frequency outside the allowable range is absolutely necessary, the team may contact the AAVC organiser directly to request a usage exemption as soon as possible. However, it will not guarantee that such a request will be granted.

Aerial Challenge Operation Objectives

The operation challenge theme set for AAVC 2026 will be focusing on the ability of the unmanned aerial system to perform precision delivery of time-sensitive, fragile-cargo using visual based landing system. However, due to the size of the operation area and the limitation of flight high (less than 20m), the objectives (landing pads) will be set so that the searching flight patterns would be dense enough to reflect an importance of the design/development consideration on flight endurance and speed of the aerial vehicle. Figure 4 illustrates the flight profile of operational challenge.

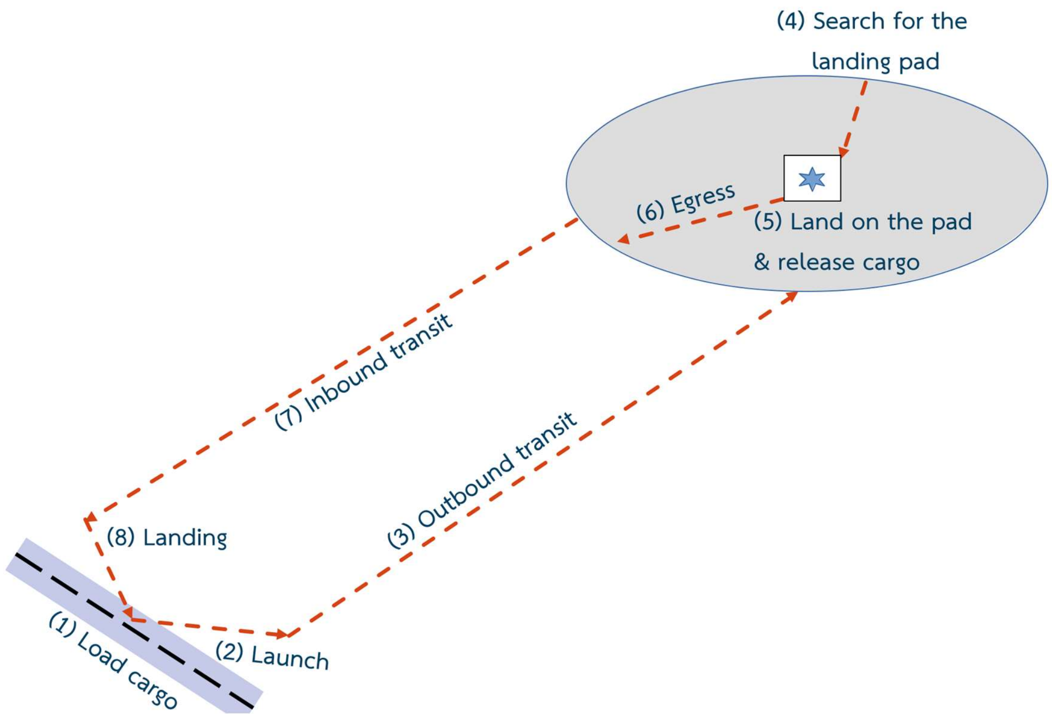

Figure 4 Operational challenge profile.

The key challenges of the operation scenario set for this AAVC2026 includes:

- Carry small fragile cargo using the team's owned specialized payload handling mechanism and carry the payload into the search area.

- Search and identify the landing pad that match the cargo's delivery assignment based on the image on the landing pad.

- Perform landing onto the "correct" landing pad and "release" the cargo gently so that its condition is intact.

- Lift-off and return to the launch and recovery point.

Please note that additional deliveries are highly encouraged In order to gain additional points. Should the team decide to do so, the aircraft shall be resupplied (cargo and/or power cells) and other payload-landing pad pairs will be assigned for the next delivery flight. This could be repeated until the cargo is delivered to all landing pads, or the operation time limit is reached. Up to four (4) landing pads will be placed across the search area.

The object representing the fragile-cargo unit used in this AAVC 2026 challenge is one large-size (no.0) raw-egg wrapped with a heart-shaped package as illustrated in figure 5

Figure 5. fragile cargo and the wrapping prepared by IAAI KMITL

The landing pad consists of a marker enclosed by a circle having its dimensions (in millmeters) as illustrated in figure 6. The marker (ArUco, Dictionary: standard-4x4), placed at the centre of each landing pad, will be generated from the open-source QR generator (https://chev.me/arucogen/) using number 1 through 6, as illustrated in figure 7.

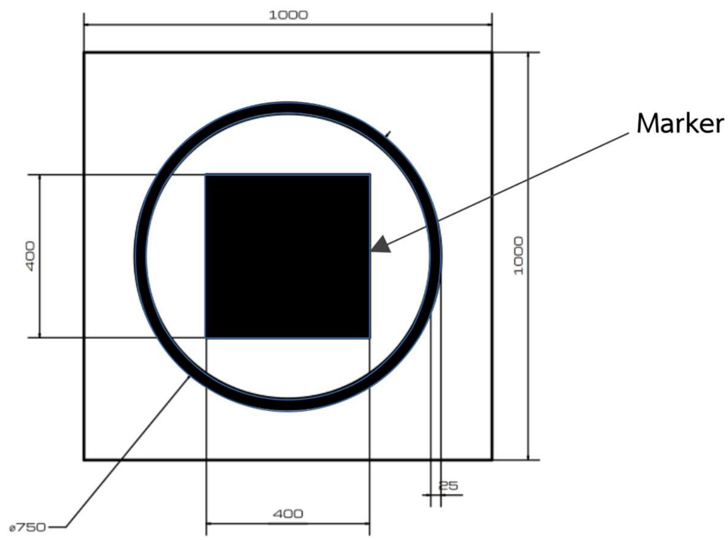

Figure 6. Geometry of the landing pad.

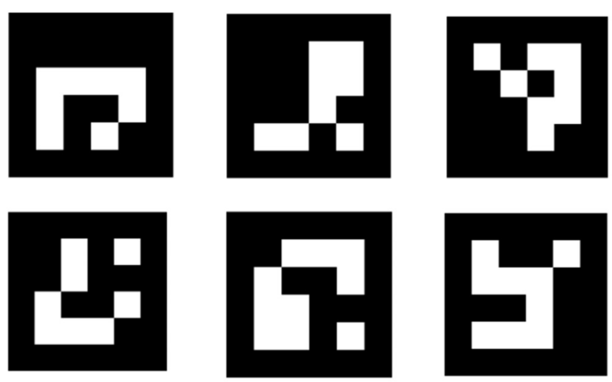

Figure 7. ArUco marker identification (1 through 6).

<u>Technical and Operational Tasks for the Participating Teams</u>

Each participating team shall develop the autonomous aerial supply delivery system using small unmanned aerial vehicles as delivery agents. Due to the limitations on the size of operating area and the urban surroundings, the vehicle must be able to perform vertical takeoff and landing. However, this does not mean that the vehicle design configuration is restricted (could be either VTOL fixed-wing, multi-rotor, helicopters, tiltrotors, tail-sitter, etc). The system may use either single or multiple vehicles to complete the task. The team could develop the system having multiple vehicles in a-cooperative manner (either using multiple vehicles of the same type for simultaneous search and deliveries or using one configuration to perform search and identify while using the other configuration solely for deliveries). However, such implementation must be done within certain technical and operational restrictions.

The teams are encouraged to conduct their studies/analyses to develop the best solution based on the operating environment parameters, operation restrictions, their current technical expertise in vehicle engineering and software engineering, and the flight performance/system capability of their current vehicle(s). The system must be developed according to the following capabilities, regardless of design configurations.

- The air vehicle(s) used in the system shall be equipped with an autonomous flight system that enables the vehicle to fly autonomously on the pre-programmed flight paths (waypoints) during operations. The flight control/management system shall be developed based on a non-proprietary, open-source hardware architecture and software stack. The team is also encouraged to develop higher level algorithms, i.e. machine learning, to enhance the autonomous flight capability including automatic target identification and localization, auto flight search pattern, and autonomous landing and cargo release.

- The system shall have a dedicated ground control station (GCS) to provide command and control for the aerial vehicles. The GCS shall consist of a primary control terminal (personal computer, fixed or mobile), a remote control for the safety pilot, a communication tower, and an external/backup power source. The system shall also be able to relay flight and imaging data from the UAS's primary control terminal to at least one external display unit for operational performance evaluation.

- The air vehicle(s) used in the system shall have an adequate energy source to complete flight operations based on the team's decision regarding the vehicle's performance, search flight patterns, operating speed, high, and loitering/hovering time over the operating area.

- The air vehicle(s) shall be equipped with the imaging system that could record and transmit imaging data to the ground control station. The imaging system could be either electro-optical (EO), infrared (IR), or a combination. The team may use a separated imaging system dedicated for visual based landing. There is no restriction on the equipment and/or technique used to acquire the position (coor-dinates) of the objective, except the minimum allowable operating high above ground.

- The air vehicle(s) used in the system shall be equipped with a payload handling system with a suitable mechanism for holding and releasing the cargo. The cargo handling mechanism can be in a form of fixed length sling/cable or other mechanical linkages/arm attached to the cargo holding mechanism e.g. claws, enclosed compartment with opening hatches, etc. If the air vehicle can carry multiple cargo, each must have its own release mechanism for independent cargo release control. There is no restriction regarding the number of payload modules installed on a single aerial vehicle as long as its maximum takeoff weight (MTOW) does not exceed the specified limit (see restrictions). The installation and operation of the payload handling system shall not exhibit unsafe behavior.

- The system shall be equipped with a telemetry system capable of transmitting data, including command and control, vehicle flight status, system operating status, video imagery, etc., between the air vehicles and the ground control station. The telemetry system shall provide at least 500 meters line-of-sight (LOS) signal coverage over the operation area while maintaining adequate signal strength during the entire operation.

The team shall perform hardware and software integration of the flight and ground control system with proper level safety in mind. This includes implementing a system engineering approach (e.g., system redundancy by design) and incorporating emergency/fail-safe measures.

Team organization

The participating team shall consist of current students from the representative institution. An institution could send multiple teams to participate. However, a member of one team cannot be listed as a member of the other team.

Each participating team shall have no more than fifteen (15) members. This includes one faculty staff member who will serve as the team advisor. Each team is allowed to have one (1) graduate student in the team. The team leader, who must not be an advisor or a graduate student, shall be the primary contact person responsible for all communications between the team and the AAVC staff throughout the operational challenge period.

Timeline and Deliverables

June 2026 : Open for registration

To register, the participants shall submit their registration by contacting at the International Academy of Aviation Industry (iaai@kmitl.ac.th) to receive registration form and preliminary report, which the team must fill-in necessary information and submit to the IAAI staff.

Please note that the participating team shall not be responsible for cost regarding registration (no registration fees). However, team shall be responsible for their accommodation and transportation from their institution to the IAAI, KMITL. Please note that due to the limited number of slots available for the aerial operation challenge, AAVC member is the priority with maximum 2 registrations per institute, the general registration will be processed on a first-come, first-served basis.

20 June 2026 : End of preliminary documentation submission deadline.

To promote proper engineering practices and ensure that the systems developed by each team can perform effectively and safely in operational challenges, each participating team must create a technical document while simultaneously developing their system after registering. This document must be submitted to the AAVC committee for evaluation.

The document, which should be a maximum of 30 pages, shall provide technical information about the engineering design and analysis of the aerial vehicle and its subsystems. The report should include evidence demonstrating that the system has sufficient flight performance and capabilities to operate with an appropriate level of safety. The contents of the technical report should consist of:

Executive summary : A summary describes the overall content of the report.

Team organization chart : The chart shall include pictures of team members and descriptions of their roles in the team, including academic staff and /or graduate students.

Concept development : The description of the thought process behind the system's development. This may include deriving system performance parameters based on the operational challenge environment, justifying the selection of vehicle configuration and system architecture, and describing the concept of operations.

Engineering Design and Analysis : The technical details of the engineering work performed during the development of the airframe structure and subsystems, as well as the implementation of the fail-safe and operating procedures. Such engineering works may include aerodynamics, propulsion, flight mechanics, system engineering, and structural analysis. The Design, Construction, and Operation Standard (DCOS) is provided as a technical guideline for airworthiness considerations in aerospace system development, e.g., ensuring the integrity of the airframe structure and subsystem (hardware and software), the aerial vehicle's flight performance, the compatibility of data-link components, and the safety of operation in both normal and emergency conditions. Appendix A outlines the content of the technical documentation, while Appendix B offers a comprehensive description of the engineering design criteria.

The CASE's technical evaluation committee will evaluate the technical reports submitted by all participants based on the technical criteria stated in the DCOS. This technical report score will be used to arrange the team's time slot during the operational challenge. Moreover, the technical evaluation score will be used as part of the overall score once the operational challenge is concluded.

20 July 2026 : End of Technical Documentation Submission deadline

28 August 2026 : Arrival, Briefing and Inspection.

All participants must arrive at the IAAI's headquarter building (Terminal 1) for registration. Registration will be open from 8:00 AM to 9:00 AM

At 9:00 AM, all participating teams shall attend the welcome meeting for the opening remarks, rules and regulations briefing, operational challenges, safety, and security matters inside the IAAI's perimeters.

After the meeting, all participating teams must present their systems to the AAVC committee for a safety inspection. The committee will evaluate each system, identify any potential safety issues, and provide feedback to the team members for correction. The teams are required to make the necessary adjustments and modifications to address these issues and bring their systems back for a recheck during the flight trial period . The technical exchange session will occur alongside the flight trials, with each participating team represented by its designated presentation team. Other participant members not involved with the flight trials are welcome to join and use this opportunity for networking and technical exchanges. Each team will have approximately 15 minutes to present, followed by a 5-minute question-and-answer period. The CASE committee will evaluate the presentation, and the evaluation scores will be used to calculate the overall score. The schedule for presentation time slots will be announced upon arrival.

Lunch will be provided for all participating team members.

28 ***August 2026 01:00-06:00 PM : Flight trial & technical exchange session***

Participating teams will have the opportunity to conduct flight trials and practice sessions to familiarize themselves with the operational area and build confidence in their system and operational readiness. The AAVC organizer will schedule trial time slots for each team on a first-come, first-served basis. Each team will be allotted 15 minutes for each trial to maximize the number of trial rotations. Teams may perform additional trial flights as long as time permits. The test site supervisor reserves the right to pause or terminate all trial activities due to weather conditions or unforeseen emergencies.

The dinner reception, along with the opening ceremony, will take place at 6:30 PM.

29-30 ***August 2026 : Flight operational challenge days.***

On the day of the flight operation challenge, all participants are required to gather at the preparation area inside the IAAI Terminal 1 building before 7:30 a.m. Each team will have their own preparation station consists of workshop tables and power outlets. Final details, including the latest weather updates and each team's operating time slot, will be announced during this time. Participants will conduct system setup and other necessary preparations while waiting for the organizing staff to transport the air vehicles and team members to the stand-by area.

At the stand-by area, the team will be taking the place of the team that has just completed their operations. While waiting for their assigned operation time slot, teams are permitted to carry out final system preparations. It is important that all preparation activities are conducted in accordance with safety rules and regulations. Teams will receive the cargo and its delivery assignment and while weighting the air vehicle to verify the takeoff weight witnessed by the CASE's field committee.

The amount of team members to be allowed inside the ground control station's operating area are as follows:

- Safety pilots: one per the number of air vehicles used in the system.
- Command and control system operators: a maximum of four members.
- Technical support personnel: a maximum of three members.

Once the team is cleared to move from the standby area to the launch and recovery area, they will have a total of twenty-five (25) minutes to perform their operation. The first five (5) minutes of this time will be dedicated to setting up the control station and positioning the air vehicle in the launch position. The team leader will notify the committee when the system is operational. If the team is able to prepare the system before the five-minute window expires, the committee will start the air operation timer, which is a twenty-minute countdown, and the remaining preparation time will be discarded. However, if the five-minute window passes and the team is still not ready, the mission timer will start automatically, resulting in a reduced operation time for the team to complete their task.

During the operation time window, the team will manage the system, oversee team activities, and make decisions within the designated ground operating area. Communication with individuals outside of the team will be limited to the CASE's field committee. Participants in the air operation challenge must adhere to the restrictions outlined by the committee; further details can be found in the regulations section.

The team leader is responsible for informing the committee about all significant activities and decisions made during the operation time window. However, the safety committee has the authority to terminate (abort) flight operations if any unforeseen circumstances arise that pose a critical safety risk.. The team leader will inform the committee of all major activities and decisions made throughout the operation time window. However, the safety committee will have the authority to terminate (abort) the flight operation in case of any unforeseen scenario that is considered critical to safety. The team would have full autonomy to operate the system, including the decision to return to the launch and recovery site in the middle of the operation should a system malfunction be detected, and restart the flight operation once the malfunction is solved. These retries could be done without penalties, as long as the operating time window does not expire. The dinner reception, along with the award and closing ceremony, will take place at 6:30 p.m. of 30 August 2026.

Operation Effectiveness Evaluation Matrix

The operational effectiveness for each team will be evaluated by the field technical committee based on the evaluation matrix detailed as follows :

Scoring Criteria	Evaluation Scheme
Take-off	Points will be given for a successful takeoff. *Additional points will be given if the takeoff is perform autonomously.*
Transit (inbound)	Points will be given for passing each transit coordinate in the ingress route order.
Landing pad Identification and localization.	Points will be given when correctly identify the landing pad that match the cargo delivery assignment. Points will also be given when the coordinate of the correct landing pad is obtained. *A significant points will be added when the identification and localization task are perform autonomously.*
Landing and release the payload.	Points will be given when the air vehicle is successfully placed the cargo onto the landing pad. Also points will be given when the air vehicle is landed onto the landing pad before release the cargo *A significant points will be added if the landing and cargo released are performed autonomously without human intervention.*
Transit (outbound)	Points will be given for passing each transit coordinate in the egress route order.
Landing	Points will be given for a successful landing within launch and recovery area. *Additional points will be given if the landing is performed autonomously.*
Condition after landing.	Points will be given if the air vehicle is returned without damage.
Cargo condition	Points will be given if the condition of the cargo is intact.
Repeating Delivery	Each flight sortie will be scored using all the criteria above. The given points for each sortie will be accumulated into the team's total points of the day.
Operation time penalty	The penalty will be represented as a "negative" points, which will be added to the overall score. Once the operation time window is reached, there is a specific penalty points for every one minute.

Restrictions

There will be restrictions that the participants must comply with to ensure operational safety and evaluation fairness throughout the challenge. Ignoring or violating such restrictions and other safety rules stated by the event organizer will incur steep penalties and may result in disqualification for the team. Key necessary restrictions are summarized as follows:

System development restrictions:

- Regardless of the vehicle design configuration and the numbers/types of air vehicles used in the system, all air vehicles' combined maximum takeoff weight (MTOW) must not exceed 25 kg.

- The team shall perform airframe assembly and subsystem integration (both hardware and software) by themselves. However, it is acceptable to have a third-party expert to supervise the assembly/integration task.

- A team may integrate their flight system hardware using the open-source flight system architecture and other stand-alone imaging and telemetry systems into the commercial-grade "empty airframe". However, the use of "complete proprietary flight system package" from commercial-grade remotely piloted aircraft manufacturers in any airframe, even if it is embedded with a custom-made high-level computing algorithm, are strictly prohibited.

- The use of any "winching" mechanism for lowering/retracting the cargo holding mechanism is not allowed. Also, it is not allowed to release multiple cargo at the same time.

Operation restrictions

- Flight line crews must always wear their personal protective equipment (PPE).

- The air vehicle must take off and land at the designated launch and recovery site.

- All flights must use the designated transit routes to enter and leave the area.

- All air operations, except takeoff and landing at a designated launch and recovery site. The flight high during transit must be strictly at 20 m. Once entering the search area, the searching flight phase shall conducted with a minimum high of 10 m AGL. The team is allowed to descent below 10 m only for the delivery on the landing pad. Maximum high in search area shall not exceed 20 m AGL.

- Activating radio control equipment and other telemetry is strictly prohibited while in the standby area.

- Leaving the operation airspace (violating geo-fencing) or entering no-fly area during the mission is strictly prohibited.

Appendix A

Guidelines for technical report structure

Section	Description	Evidences
Executive summary (10 points)	The one-page summary of the entire report.	
Team organization (10 points)	The information describing the team organization and the role / function of each team member, including the academic staff advisor and graduate student member (if any).	The team organization structure could be represented in various forms such as : - Organization chart - Table describing team member's roles and functions. * Picture of team member (grouped or individual) is recommended*
Concept development (30 points)	The thought process used in developing the aerial vehicles and associated subsystems as well as the operational concept of such systems.	- Derivation of the delivery system capability parameters (air vehicles and subsystem performances) captured from the operational challenge environment and scenario. - Description of the selected delivery system (air vehicle configuration and overall system architecture) concept, including the rational used to justify the decision for such concept. - Description of the concept of operation of the system (how such system would be operates in order to fulfill the operation challenge objectives). This may include designed mission profile, target search and package delivery tactics, etc.)

Guidelines for technical report structure (Cont.)

Outline	Detailed Arguments	Means of Evidences
Engineering design and analysis (40 points)	The technical detail regarding engineering work applied during the development of the aerial vehicle(s) and subsystems.	- Description of the method and assumptions used in engineering analysis to ensure the integrity of the air vehicle's airframe structure during flight, e.g., flight loading analysis, airframe structural design and analysis, and/or structural testing. - Description of the method and assumptions used to verify the flight performance of aerial vehicles according to the derived capability stated in the Concept Development section, e.g., aerodynamics, propulsion, and flight performance analysis. - Description of the method and assumptions used to ensure the integrity of the subsystem (hardware and software) during flight operation under operating challenge environment. This could be in the form of the description of system design architecture with detailed software functions (operating modes, safety features, etc.) - Description of procedure used for operating the system to ensure that the system will operate with proper precautions covering normal and emergency operation conditions. - Description of the communication system specification (frequency and power) to ensure the data-link system compatibility (frequency & power) with the operational environment scenario.

Appendix B

Design, Construction, and Operation Standard

(DCOS)

For autonomous aerial vehicle participating in AAVC

Disclaimer:

- The criteria in this DCOS will serve as engineering guidelines for the participating teams, promoting the use of proper engineering practice in the development of aerospace systems according to airworthiness standards.

- The criteria stated in this DCOS (design aspect) will also be used to evaluate the technical documents submitted by the participating teams and the readiness status of the aerial vehicles in the safety inspection.

- While the criteria in this DCOS are not meant to be fully enforced, any aspect of the system that leads to operational hazards will not be ignored. Upon the safety inspection, the AAVC committee will report all potential safety issues to the participating teams for correction. Unless such issues are resolved to ensure an adequate level of safety, the AAVC committee shall hold the rights to prohibit the vehicle from flight operations.

The criteria in DCOS are as follows :

- The aerial vehicles shall achieve the flight performance that meets or exceeds the design capability parameters for all possible load carrying configurations and flight conditions without exhibiting unsafe behavior due to aerodynamics, power, and/or control authority issues.

- The limitation of flight performance, e.g., minimum speed, turn rate, turn radius, load factor, center of mass, etc., must be clearly defined and, if applicable, implemented into a flight system as envelop protection.

- The vehicle's propulsion system shall have a reasonable safety margin regarding excess thrust/power to ensure that the components do not reach their limits during normal operations for all possible load carrying configurations and flight conditions.

- The vehicle's power source shall provide sufficient power for propulsion and subsystems under all possible load-carrying configurations and flight conditions, as well as an adequate safety margin (reserve capacity) to prevent unintentional power drop due to energy starvation.

- The structure and assembly of the airframe components shall withstand the loads (flight, ground operation, and transportation) under all possible flight conditions and load configurations without exhibiting unsafe behavior and/or premature damage.

- The system architecture, including air and ground elements, shall be designed with a suitable degree of redundancy, such as backup power, redundant sensors, etc.

- The system component hardware used in the system, both air and ground elements, shall be compatible in terms of performance, communication protocols, connections, and power characteristics to ensure operational safety and reliability.

- The system components' hardware on aerial vehicles shall be securely attached to the vehicle's airframe using the proper fastener and/or retaining hardware to prevent movement, which could cause weight shift and/or damage to nearby components during operations.

- The moving mechanism (e.g., payload handling) installed onto the airframe shall be designed with sufficient rigidity and properly use hardware components to prevent collision /interference with nearby components due to deformation. The detachable payload (if present) shall be securely installed on the payload handling mechanism using proper installation hardware to prevent unintentional release during flight.

- The proper installation clearance between rotating parts/moving mechanisms shall be considered to prevent collision or interference with other components due to structural deformations during operation.

- The wiring connection between components must be secure using proper types of connectors and wire gauges according to the signal and/or electrical loads requirements.

- Critical system components and wiring shall be adequately protected from the outside environment, such as dust and rain, to prevent physical damage and/or short circuits.

- The system shall implement envelope protection to prevent users from operating outside the system limits.

- The system shall implement some form of fail-safe operating mode for recovering/redirecting the air vehicles to a safe location in case of emergency. Such mode(s) could be activated at any time during operation.

- Both normal and emergency operating procedures shall be clearly established, considering all possible scenarios that may occur during operations.

- Based on their roles, the designate operators shall be assigned proper personal protective equipment (PPE), such as protective glasses, safety helmets, and reflective vests.